

A global mapping technique for GPS-derived ionospheric total electron content measurements

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Abstract. A worldwide network of receivers tracking the transmissions of Global Positioning System (GPS) satellites represents a new source of ionospheric data that is globally distributed and continuously available. We describe a technique for retrieving the global distribution of vertical total electron content (TEC) from GPS-based measurements. The approach is based on interpolating TEC within triangular tiles that tessellate the ionosphere modeled as a thin spherical shell. The high spatial resolution of pixel-based methods, where widely separated regions can be retrieved independently of each other, is combined with the efficient retrieval of gradients characteristic of polynomial fitting. TEC predictions from climatological models are incorporated as simulated data to bridge significant gaps between measurements. Time sequences of global TEC maps are formed by incrementally updating the most recent retrieval with the newest data as it becomes available. This Kalman filtering approach smooths the maps in time, and provides time-resolved covariance information, useful for mapping the formal error of each global TEC retrieval. Preliminary comparisons with independent vertical TEC data, available from the TOPEX dual-frequency altimeter, suggest that the maps can accurately reproduce spatial and temporal ionospheric variations over latitudes ranging from equatorial to about $\pm 65^\circ$.

1. Introduction

Data collected from a continuously operating global network of Global Positioning System (GPS) receivers have been used to produce global maps of ionospheric total electron content (TEC) at sub-hourly intervals. These maps can be used for calibrating the delay of radio signals due to the ionosphere or for total electron content studies. Since the GPS observables are generated continuously and are available over the Internet typically within 24 hours or less after reception, maps could be produced continuously on a subdaily basis and used for estimating vertical total electron content over any area of the globe, possibly supplanting or enhancing the climatological approaches that have been used in the past.

The global GPS network (GGN) consists of more than 100 dual-frequency geodetic quality receivers operated by several agencies and coordinated by the International GPS and Geodynamics Service, or IGS [Zumberge *et al.*, 1994]. Each GPS receiver in the global network tracks between four and 12 satellites simultaneously over a site. There have been several reported techniques for mapping TEC measurements in the vicinity of a dual-frequency GPS receiver [Lanyi and Roth, 1988; Coco *et al.*, 1991; Coster *et al.*, 1992; Sardon *et al.*, 1994]. In this paper we describe a global TEC mapping technique that has greater time resolution than a previously reported approach based on surface harmonic fits to IGS data over several-hour periods [Wilson *et al.*, 1995]. Preliminary comparisons between the maps and independent TEC data from the TOPEX dual-frequency ocean altimeter suggest that the mapping technique can accurately retrieve large-scale (~ 500 km) features in the ionosphere over a wide latitude range (TOPEX data are available up to latitudes of about $\pm 67^\circ$).

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The purpose of this paper is to describe the global mapping technique and show preliminary comparisons with TOPEX measurements. We will also discuss how the retrieval algorithms have been extended to include estimation of hardware interfrequency biases that can corrupt GPS measurements of ionospheric delay. First, we will summarize how GPS data are processed to derive slant line-of-sight (LOS) TEC measurements.

2. Ionospheric Measurements From the GPS Global Network

In this section, we will briefly review how TEC measurements are derived from GPS observables. The general approach of combining GPS pseudorange and phase measurements to extract TEC has already been described in the literature [Lanyi and Roth, 1988], and in this section we will emphasize enhancements that have been applied in our processing of global network data.

The GPS receivers in the global network generate two delay observables for each satellite being tracked: pseudorange delay (P) and carrier phase delay (Φ). Each observable is acquired at the two GPS transmission frequencies: f_1 (1.5754 GHz) and f_2 (1.2276 GHz). The frequency-differenced phase delays (phase measurements converted to equivalent ranges) provide very precise measurements of TEC changes but contain an overall bias associated with integer cycle ambiguities. The frequency-differenced pseudorange data provide an absolute measure of the total ionospheric delay between satellite and receiver but have significantly more multipath and system noise than the phase-based data. In addition, the pseudorange delays are biased by interfrequency delays in the satellite and receiver hardware. These hardware-dependent biases can be constant over several days [Wilson and Mannucci 1993; Coco et al., 1991] and can be calibrated as part of the TEC mapping process (see section 3.3; other calibration methods are also possible).

A TEC observable is formed from the frequency-differenced phase delay by adjusting its level based on the pseudorange delay. A single leveling constant is determined for each set of points in a time series of satellite-receiver data that share a common phase bias, a so-called "phase-connected" arc of data. Let N be the number of measurements in a phase-connected arc for a given receiver r and satellite s .

For each datum i , the pseudorange delays are denoted by P_{1i}^{rs} and P_{2i}^{rs} for frequencies f_1 and f_2 , respectively; the corresponding phase ranges are Φ_{1i}^{rs} and Φ_{2i}^{rs} . The leveling bias B_{rs} is computed as the average difference between ionospheric delays computed from the pseudorange and phase, summed over the arc, and weighted by an estimate of the pseudorange noise (phase noise is negligible by comparison):

$$B_{rs} = \frac{\sum_{i=1}^N \frac{1}{\sigma_i^2} \{ (P_{2i}^{rs} - P_{1i}^{rs}) - (\Phi_{1i}^{rs} - \Phi_{2i}^{rs}) \}}{\sum_{i=1}^N (1/\sigma_i^2)} \quad (1)$$

where σ_i is an estimate of the standard error of the frequency-differenced pseudorange noise, which can depend on elevation angle, receiver type, multipath environment, and tracking mode of the receiver (precision code or codeless tracking). The TEC observable for each measurement i is the phase-based ionospheric delay $\Phi_{1i}^{rs} - \Phi_{2i}^{rs}$ added to the above bias B_{rs} . One nanosecond of delay corresponds to about 2.85×10^{16} el/m², or 2.85 TECU.

Equation (1) is a formula for the mean difference between phase and pseudorange ionospheric observables, assuming unequal standard errors for each point [Bevington, 1969]. The pseudorange errors have been estimated from an analysis of GPS global network data by computing the mean difference of pseudorange and phase (ionospheric combination, after leveling) in 5° elevation angle bins, averaged over all receivers of a similar type [Runge et al., 1995]. The weighting factors σ_i currently used in computing the TEC are shown in Figure 1. These values apply particularly to Allen Osborne Associates Rogue and TurboRogue receivers that populate the IGS network, covering both code and codeless tracking modes, and may be different for other receivers. Less precise codeless tracking is necessary since encryption of the GPS ranging code to prevent spoofing (applied to most satellites since January 1, 1994) requires a cross-correlation tracking mode that overcomes ignorance of the precise ranging code. This results in significantly increased system noise.

Phase noise is generally less than 0.1 TECU (5-minute integration time) and so is negligible compared with pseudorange noise. The precision of the pseudorange is primarily limited by two factors: thermal noise and multipath (interference from indirect reflected signals that coherently add to the direct signal). Both factors can increase in codeless mode.

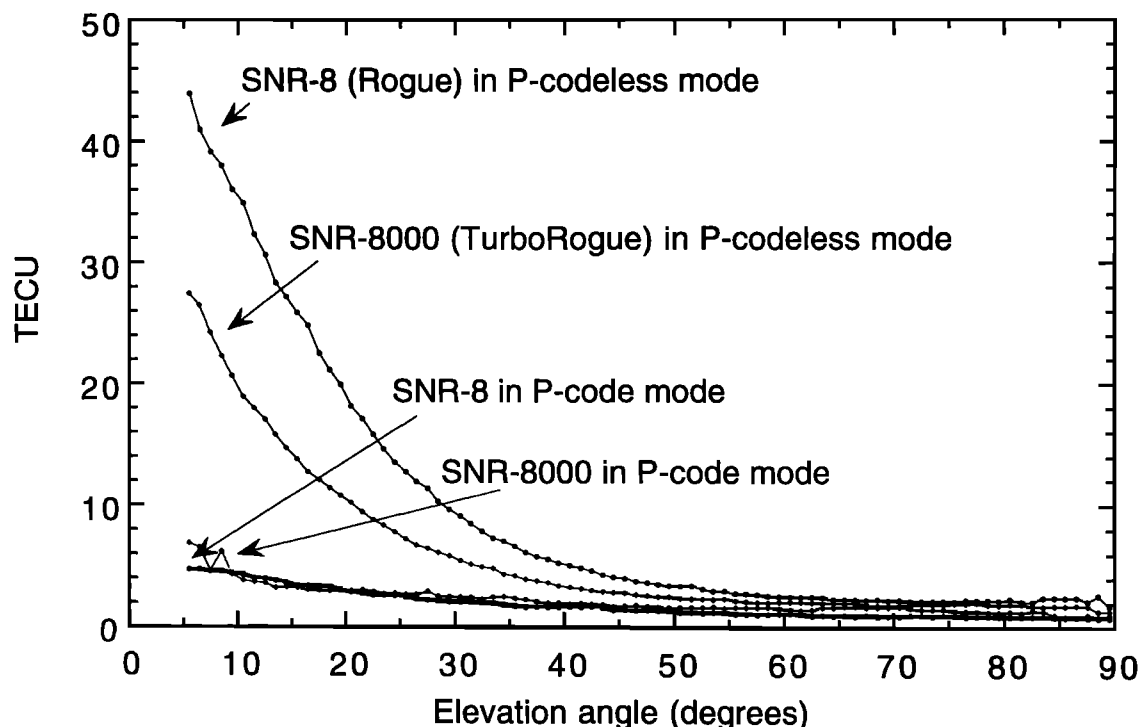


Figure 1. The weighting factors σ_i used to compute the carrier phase level for the ionospheric combination (equation (1)). These curves represent elevation-dependent pseudorange noise for four separate receiver/tracking mode scenarios and 30-sec integration times.

System noise is generally constant among receivers of the same type; multipath noise varies with the electromagnetic environment at each receiver. Therefore the curves in Figure 1 represent an average of multipath noise levels for global network sites, which are generally configured to minimize multipath.

A data preprocessing step is required before TEC can be derived in order to identify possible phase breaks in either the $L1$ or $L2$ phase measurements [Blewitt, 1990]. An automated data-editing algorithm is necessary since in excess of one million data points are typically available from the global network every 24 hours. The editing algorithm can occasionally correct cycle slips in the phase data and identify outliers, but most important, it is a robust algorithm for detecting them. After editing, the data are divided into sets of phase-connected arcs for the processing as described above.

A minimum elevation cutoff of 10° is applied to avoid processing particularly noisy measurements (see Figure 1). Data from arcs less than 20 min long are rejected because leveling accuracy depends the number of points within a phase-connected arc. Af-

ter the leveling, the RMS difference between pseudorange and carrier phase measurements is computed and compared with the expected value based on the average noise data (Figure 1). If the pseudorange scatter exceeds the expected level by a factor of 3, the arc is rejected. This criterion has occasionally edited out most of the data from sites where multipath noise is much worse than average.

3. Global Mapping of Ionospheric Measurements Derived from GPS

The distribution of receivers in the GPS global network as of late 1996 is shown in Figure 2. The circle around each site represents the area within which measurements of TEC can occur, assuming a fictitious ionospheric shell at a height of 450 km and a 10° elevation mask. Within each circle, between four and 12 simultaneous TEC measurements are available at a single time. These measurements, obtained every 30 s, are interpolated in both space and time to produce the global TEC maps.

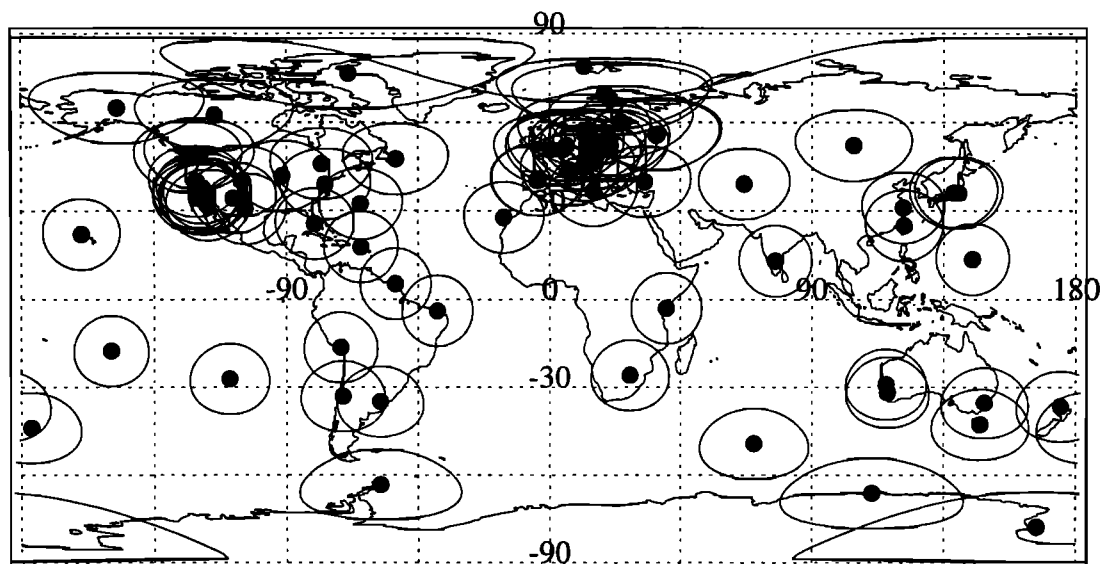


Figure 2. The distribution of Global Positioning System (GPS) global network sites as of late 1996. Typically, six to eight GPS receivers are simultaneously tracked by each receiver. The circle above each site represents the ionospheric shell area where the measurement pierce points will lie, assuming a shell altitude of 450 km and a minimum elevation angle of 10° .

The maps of vertical TEC are produced from the line of sight GPS observations using an ionospheric shell model assumption. The slant TEC measurements, typically decimated to a 5-min sampling rate, are used to estimate a vertical TEC value at the location of a fictitious “ionospheric pierce point” (IPP), defined by the intersection of the satellite-receiver line of sight with a thin spherical shell surrounding the Earth at a height of 450 km. The typically six to eight estimates of vertical TEC from each satellite-receiver line-of-sight at a given epoch represent a sampling of the Earth’s vertical TEC distribution at the pierce point coordinates. This sampled distribution is then interpolated to produce spatially continuous vertical TEC maps. In this section, we describe in detail a method for performing the interpolation, both spatially and temporally.

3.1. Scaling Slant Observations to Equivalent Vertical TEC Measurements

GPS data provide a measurement of the total column density of electrons along the satellite-receiver ray path. The ionospheric electron density varies along the line of sight and is most significant between

altitudes of about 150 km (*E* layer) and several thousand kilometers (*F* layer and plasmasphere), with peak densities generally occurring at altitudes between 250 and 600 km. Vertical TEC at the fictitious pierce points is estimated assuming the electron density profile is constant along the ray path. For any fixed vertical density distribution, the slant measurements can be scaled to equivalent vertical TEC measurements using an elevation-dependent factor.

The distribution used to compute the scaling factor is a constant-density slab with extended tails in the topside and bottomside regions [Coster *et al.*, 1992]. Figure 3 shows the distribution along with the scaling factor as a function of elevation angle. This distribution was used in an accuracy study of scaling functions with different elevation angle behavior. On the basis of comparisons with vertical TEC measurements from an independent instrument (from the TOPEX satellite, discussed below), the tapered slab function of Figure 3 produced the best global average agreement between TOPEX and GPS maps [Yuan *et al.*, 1995]. The shell height for the pierce point computation is 450 km, roughly the median height of electron density for typical daytime profiles. A significantly lower height (e.g., 350 km) reduces the

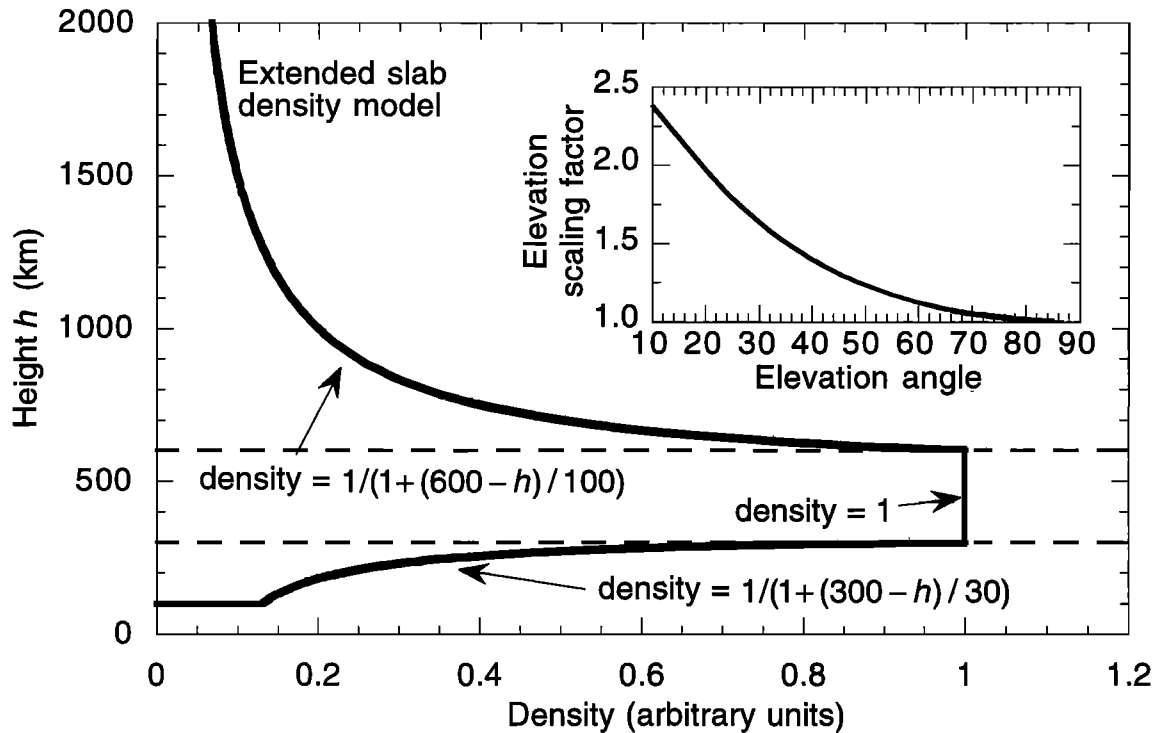


Figure 3. Electron density profile for extended slab model, from which the elevation scaling function is derived (for details, see *Coster et al.* [1992]). Inset shows the scaling factor as a function of elevation angle.

pierce point separation above the receiver, which appears to reduce accuracy.

The vertical electron density distribution is generally not constant along the ray path and is distributed over several thousand kilometers in altitude. It has been estimated that vertical TEC can be in error by 10-20% when an elevation-dependent scaling factor is used at low elevation angles (5° - 10°), and in regions of significant TEC gradients, for example at low latitudes or at dusk local times [*Klobuchar et al.*, 1993; *Tsedilina and Weitsman*, 1992]. However, the maps are not necessarily inaccurate to that degree since the larger error measurements are combined with high-elevation measurements where the vertical scaling errors are much smaller. Comparisons with independent TEC measurements bear this out: In regions where vertical scaling errors are expected to be significant, the maps are more accurate than worst-case estimates of elevation scaling errors (3 TECU accuracies are typical near GPS receivers [see *Mannucci et al.*, 1994; *Ho et al.*, 1997]. One TECU = 10^{16} el/m 2).

3.2. Interpolating the Measurements Globally Over the Ionospheric Shell

Using the shell model, a unique shell intersection point can be defined for each GPS measurement. The global maps are formed by interpolating vertical TEC estimates between the pierce points. In this section we will define how the spatial interpolation is performed. The general method can be described as follows: A set of basis functions is defined whose domain is the spherical ionospheric shell. The global maps consist of a linear combination of these functions, and the multiplicative coefficients are computed based on minimizing the residual difference between the data and maps evaluated at the measurement pierce points. The vector of best fit coefficients combined with the basis functions defines a global TEC distribution that provides interpolated values covering the entire sphere.

3.2.1. Defining basis functions for interpolation. An important feature of the basis set we have constructed is "local support": Each basis func-

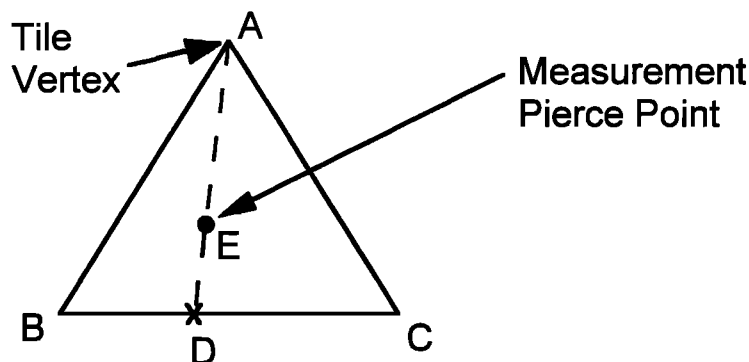


Figure 4. Illustration of the geometry for determining the total electron content (TEC) within a tile. A great circle arc passing through vertex *A* (chosen arbitrarily) and the measurement pierce point *E* intersects the side opposite *A* at point *D*. The TEC at point *D* is determined by linearly interpolating between the TEC at vertices *B* and *C*. Then, the TEC is interpolated to point *E* based on the TEC at points *A* and *D*.

tion is nonzero over a limited portion of the sphere, so each data point affects the TEC retrieval over a limited neighboring region. This locality is desirable on physical grounds since dynamical changes in the ionosphere over widely separated global regions are generally uncorrelated.

We have constructed locally supported basis functions by interpolating TEC within individual triangular tiles that tessellate the sphere. Each tile can be regarded as a rigid plate that tilts to follow the measured TEC gradients within the tile. The rigidity of the tiles implies that the global TEC distribution is approximated by piecewise bilinear TEC gradients. The three vertices of each tile are separated by roughly 800 km; different vertex spacing is easily accommodated with our method.

The intratile TEC for our rigid plate model can be expressed as a linear combination of the TEC values at each vertex of the triangular tile. Each vertex TEC value is multiplied by a weighting function that depends on the distances to the three vertices of the intersected tile. Within the formalism of the least squares estimation, the weighting functions are the locally supported basis set, and the vertex TEC values are the estimated coefficients.

Constructing the weighting functions is a two-step process. In the first step (see Figure 4), a great circle arc is constructed that passes through vertex *A* (chosen arbitrarily) and the measurement pierce point *E* located at (θ, ϕ) . This arc intersects the other side of the spherical triangle at point *D*. Once the location of *D* is determined, the TEC at point *D* is

derived from distance-weighted interpolation of the TEC values at vertices *B* and *C*:

$$V_D = \frac{DC}{BC} V_B + \frac{BD}{BC} V_C \quad (2)$$

where *BD* is the arc length between points *B* and *D* and V_A is the TEC at vertex *A*, etc. In the second step, the TEC is similarly interpolated between points *A* and *D* to measurement pierce point *E*:

$$V_E = \frac{ED}{AD} V_A + \frac{AE}{AD} V_D \quad (3)$$

Substituting the expression for V_D (equation 2) into equation (3) yields

$$V_E = \frac{ED}{AD} V_A + \frac{AE}{AD} \left[\frac{DC}{BC} V_B + \frac{BD}{BC} V_C \right] \quad (4)$$

Let T_{rs} represent the slant range delay between a receiver and satellite, which pierces the ionospheric shell at latitude and longitude (θ, ϕ) . On the basis of equation (4), we can write

$$T_{rs} = M(E) \sum_{i=A,B,C} W(\theta, \phi, i) V_i \quad (5)$$

where $M(E)$ is the elevation scaling function, V_i is the value of TEC at vertex *i*, and $W(\theta, \phi, i)$ are weighting functions for a shell intersect point having latitude and longitude (θ, ϕ) . This equation implicitly defines the basis set (the W functions) for the interpolation, which can be derived by inspection of equation (4). The W functions are nonzero in a lim-

ited region because only those functions associated with the intersected tile are involved in the sum of equation (5). The TECs at the grid points serve as the coefficients of the basis set and are the estimated parameters in the least squares retrieval.

Using bilinear spline functions for representing the TEC distribution should, in theory, produce higher accuracy retrievals than more conventional “pixel-based” techniques used in tomographic approaches [Juan *et al.*, 1997], assuming a similar number of estimated parameters. In general, global TEC distributions can be accurately represented by linear changes in TEC over tile-sized regions (see Figures 6 and 7). Using a similar number of pixels of constant density results in a less realistic ionospheric model; with such a basis set, the retrieved map would probably exhibit significant discontinuities at pixel boundaries.

The construction of the triangular tiling is performed as follows. First, an icosahedron is projected onto the sphere, aligned such that two of the 12 vertices are at the north and south poles and one of the nonpolar vertices is aligned with the prime meridian. The 12 vertices define 20 equilateral spherical triangles on the sphere. Finer tiling is achieved by successive subdivisions of these spherical triangles. At each step, a given triangle is subdivided into four smaller triangles by bisecting each of the sides of the triangle and connecting the new vertices. The triangular grid structure used most commonly for the global maps halts the subdivision process at a granularity of 1280 spherical triangles or tiles, corresponding to 642 vertices.

3.2.2. Reference frame for the ionospheric shell. Even with the extensive global distribution of GPS receivers that is currently available, at any instant there exist regions several thousand kilometers in extent that are not sampled by measurements (see Figure 2). We have used a nearly Sun-fixed or local time reference frame for the ionospheric shell which helps maintain interpolation accuracy in the presence of data gaps.

In a Sun-fixed frame, if the longitudinal distance between receivers is large, several hours may elapse between successive measurements at a particular shell location (local time). Interpolation accuracy therefore depends on the stability of the ionosphere in a local time reference frame, which is generally less variable than in an Earth-fixed frame. Above an Earth-fixed point, the most significant TEC variation is typically due to the changing local time, causing

diurnal TEC variations which peak in the afternoon and are a minimum at night. Since this dominant mode is “frozen” in a local time reference frame, the TEC at a fixed local time will tend to be more stable than the TEC at a fixed longitude.

The ionospheric plasma is produced primarily by photoionization, while its dynamics is strongly affected by the fact that ionospheric electrons are constrained to follow the geomagnetic field lines (except at very high latitudes). Degree of photoionization is strongly correlated with local time, hence the advantages of a local time reference frame discussed above. Furthermore, we expect that the TEC distribution may show less variation in a geomagnetic latitude reference frame rather than geographic latitude. An orthogonal coordinate system, where every vertex is fixed in both local time and geomagnetic latitude, cannot be defined since the magnetic pole is not aligned with the rotation axis of the Earth. Therefore a reference frame has been defined that is as nearly Sun fixed as possible, based on the “solar magnetic” system described by *Knecht and Shuman* [1985]. The Z axis of these coordinates is along the geomagnetic pole, chosen to intersect the Earth’s surface at 78.5° N and 291.0° E. The Y axis is perpendicular to both the pole and the Sun-Earth (local noon) direction. The X axis, which defines the zero of longitude, wobbles back and forth by $\pm 11.5^\circ$ once every 24 hours.

3.3. Estimating Calibrations for Hardware Interfrequency Biases

Ionosphere measurements derived from differencing dual-frequency GPS delays include instrumental biases in both the receiver and GPS satellite transmitters [Lanyi and Roth, 1988]. The instrumental bias is caused by differential delays between the f_1 and f_2 signal processing paths introduced by the analog hardware, and are generally in the range ± 9 TECU for the satellites and up to ± 30 TECU for Allen Osborne Associates SNR-8 (Rogue) and SNR-8000 (TurboRogue) receivers [Wilson *et al.*, 1995; Wilson and Mannucci, 1993]. Accurate TEC measurements and maps require calibration of these biases, which are estimated simultaneously with the TEC maps. Each LOS measurement of TEC is assumed to depend on the biases as well as the vertical TEC distribution represented by the triangular grid; additional bias terms are introduced into equation (5) as follows:

$$T_{rs} = M(E) \sum_{i=v_A, v_B, v_C} W(\theta, \phi, i) V_i + b_r + b_s \quad (6)$$

where b_s and b_r are the instrumental delays in the satellite transmitter and GPS receiver, respectively.

Solving for the biases simultaneously with the vertex TEC is possible because these unknowns have different time and elevation-angle dependencies. The bias contribution to the measured delay is independent of elevation angle and is assumed constant over an entire day (an assumption which appears correct for Rogue and TurboRogue receivers). The contribution to the delay from TEC changes with elevation angle, allowing the vertex TEC estimates V_i to be retrieved independently from the biases in the fitting. Day-to-day repeatability of the bias estimates at the 1 to 3-TECU level has been reported by several groups that use a similar technique for bias estimation from GPS data [Coco et al., 1991; Wilson and Mannucci, 1993, 1994; Sardon et al., 1994].

3.4. Time Dependence of the TEC Maps

The global TEC distribution is time-varying, and therefore the vertex TEC values are periodically reestimated, every 15 min to 1 hour (the update rate could be as often as the data sampling rate). Each updated map represents the global TEC distribution in the solar-magnetic reference frame, time-averaged over the update interval. In this section, we describe how a series of global maps are formed sequentially in time.

A factorized Kalman filter algorithm [Analytical Sciences Corporation, 1989; Bierman, 1977] is used to produce updated least squares fits sequentially for each interval: The batch of data from the latest interval is combined with previously solved-for vertex TEC values to produce updated least squares estimates. These updated coefficients then become the prior estimates for the next interval, and so on. Before data processing begins, the vertex TEC are initialized using a climatological model [Llewellyn and Bent, 1973]. Other models capable of predicting TEC have also been implemented [Bilitza et al., 1988; Daniell et al., 1995].

The Kalman filter algorithm limits the influence of past measurements on the most recent fit. Formally, this is accomplished by increasing the variances of the fitted parameters prior to each measurement update. The estimated parameters are modeled as stochastic, describing a "random walk" in time,

an approximation that has also been used for estimates of tropospheric delay using GPS [see Lichten and Border, 1987], and later for ionospheric mapping as well, by Mannucci et al. [1993] and Sardon et al. [1994]. The difference between successive realizations of a random walk process has zero mean, with a variance that increases linearly in time. Therefore, in the absence of measurement updates, the latest vertex TEC estimates remain constant in the solar-magnetic frame. The uncertainty of the estimates increases with time according to the following equation [Lichten, 1990]:

$$\Delta\sigma^2 = \dot{q}\Delta t \quad (7)$$

where $\Delta\sigma^2$ is the variance increase of the vertex TEC estimate, $\dot{q}$ is a constant variance rate (TECU^2/h), and Δt is the time between map updates (e.g., 30 min).

The rate at which the variance of the TEC estimates increases with time determines how rapidly the influence of older maps is "forgotten" compared with the newest data. Maps have been generated with standard deviation rates ranging from $0.7 \text{ TECU}/\text{h}^{1/2}$ to $8.4 \text{ TECU}/\text{h}^{1/2}$. Generally, better results have been obtained using the smaller rates, although actual ionospheric variations, particularly in the highly dynamic low latitudes, can exceed 10-20 TECU in a 2 hour period. Small variance rates increase the correlation between successive maps, which increases the temporal smoothness of the maps, and appears to minimize temporal "jumps" or discontinuities in the fitted coefficients.

The covariance matrix of the fitted coefficients can be used to produce a "formal error map" associated with each TEC map. The formal error $E(\theta, \phi)$ is an estimate of the standard deviation of the retrieved TEC for a point at (θ, ϕ) , computed as follows:

$$E(\theta, \phi) = \sum_{i,j=1}^3 W(\theta, \phi, i) W(\theta, \phi, j) P_{ij} \quad (8)$$

where $W(\theta, \phi, i)$ is the weighting function for vertex i used in the interpolation (see equation (5)) and P_{ij} is the covariance between vertex TEC parameters i and j :

$$P_{ij} \equiv \langle (V_i - \bar{V}_i)(V_j - \bar{V}_j) \rangle \quad (9)$$

where angle brackets denote the expectation value and $\bar{V}_i$ is the mean value of vertex V_i , etc. The full covariance matrix is estimated in the Kalman filter

along with the state [Analytical Sciences Corporation, 1989]. Derivation of the formal error equation (8) is straightforward, using the TEC model equation (5) and the definition of standard deviation of a random variable.

The formal error depends on the level of measurement noise, the geometrical distribution of the measurements, and the time that has elapsed since the last measurement update. Formal error maps are most useful for determining areas where the TEC estimates are of reduced precision due to lack of measurements. However, the formal error is generally not used as a measure of accuracy for two reasons: It does not account for errors in the maps due to mis-modeling (e.g., errors resulting from a shell model of the ionosphere), and it depends on the process noise variance, which is an input parameter that may not reflect actual ionospheric variability. In section 4, we discuss a method for estimating the actual accuracy of the maps.

3.5. Combining Data With Climatological Models

The data-driven mapping technique described in this paper also uses information from global climatological ionosphere models developed over the last several decades [Llewellyn and Bent, 1973; Bilitza et al., 1988]. This produces robust retrievals in the presence of large data gaps, or in the event data outages occur when retrieving data from far-flung global sites. For example, unavailable data from two or three critical sites in the southern hemisphere or low latitudes (see Figure 2) could produce longitudinal data gaps of 5000 km or more. The retrieval may be suspect in these remote areas, even assuming the ionosphere is fairly constant in local time. From a practical point of view, a global mapping technique that maintains a reasonable global accuracy, no matter what the precise data scenario, should be useful.

The climatological model information is included as simulated data, either before the actual data are processed, or periodically between successive data batches, for example hourly. In both cases, the model is used to simulate vertical TEC measurements with pierce points at each vertex location. The influence of the simulated measurements on the fit can be carefully controlled because the parameter estimation procedure uses a weighted least squares approach [Analytical Sciences Corporation, 1989]. A standard deviation of 10 TECU is assumed for the

simulated climatological “data,” while GPS data are generally weighted with standard deviations in the range 0.5–2 TECU (depending on arc length and elevation angle range).

Vertex TEC values that have not been updated for several hours due to coverage gaps, or that were never in the proximity of data, are most strongly influenced by the model information. In data-rich regions (such as over Europe and North America) the model information has virtually no effect on the retrieval since the combined information from all the measurements has significantly more weight than the climatological information.

Standard deviations in the range 5–10 TECU for the simulated data tend to produce the best agreement with TOPEX, for comparisons at medium to solar minimum conditions, although the exact weighting is not critical. However, a standard deviation much larger than 15 TECU negates the additional benefit of regularizing the maps: Adding uniformly distributed climatological data tends to smooth the maps spatially compared with completely data driven, unconstrained retrievals [see Ho et al., 1997].

4. Results

An example of a global ionospheric map (GIM) is shown in Plate 1, for February 9, 1995. False color is used to represent vertical TEC values over a latitude range extending from -80° to $+80^\circ$, covering all longitudes, during the period 1600–1630 UT. The locations of the GPS receivers used in the solution are denoted by filled circles.

The TEC distribution is plotted in a Sun-fixed, or local time, coordinate system. This system, in which the latitudes are geographic and each longitude corresponds to a fixed local time, differs from the solar-magnetic coordinates (section 3.2.2) in which the grid vertices are fixed. However, Sun-fixed is a convenient system for displaying how the TEC features are correlated with continental boundaries.

The TEC map shows expected global-scale features: the global maximum TEC near the equator and the diurnal variation of TEC that peaks about 1400 LT, decaying to a minimum just prior to sunrise. The double-hump structure characteristic of the equatorial anomaly appears most clearly between 1000 and 1300 LT. The overall structure of the central low-latitude enhancement is curved with respect to fixed geographic latitude loci, possibly due to the influence of the geomagnetic field. Loci of constant

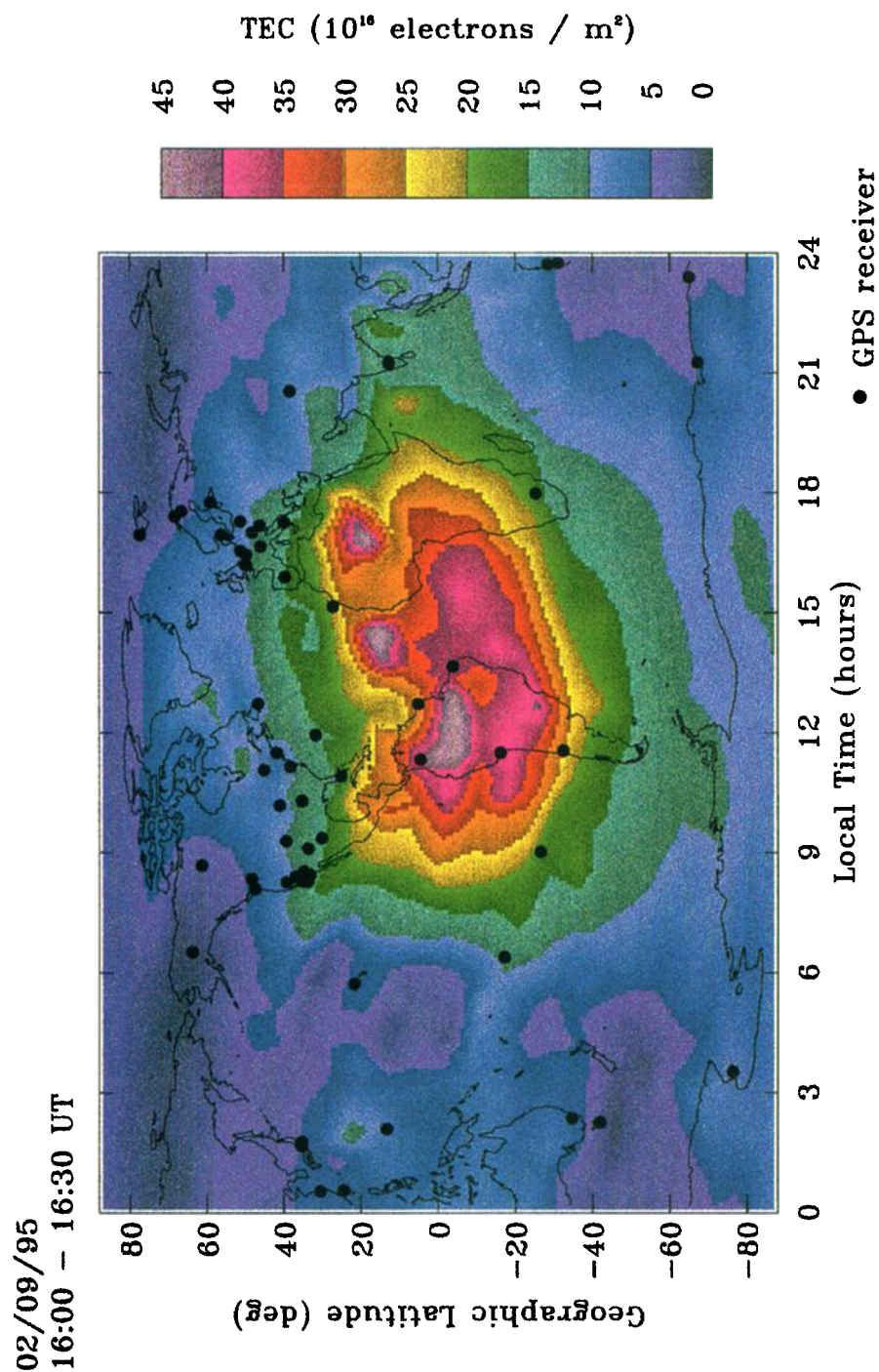


Plate 1. Global ionospheric TEC map for February 9, 1995, covering the time period 1600–1630 UT. Vertical TEC is plotted in false color as a function of geographic latitude and local time.

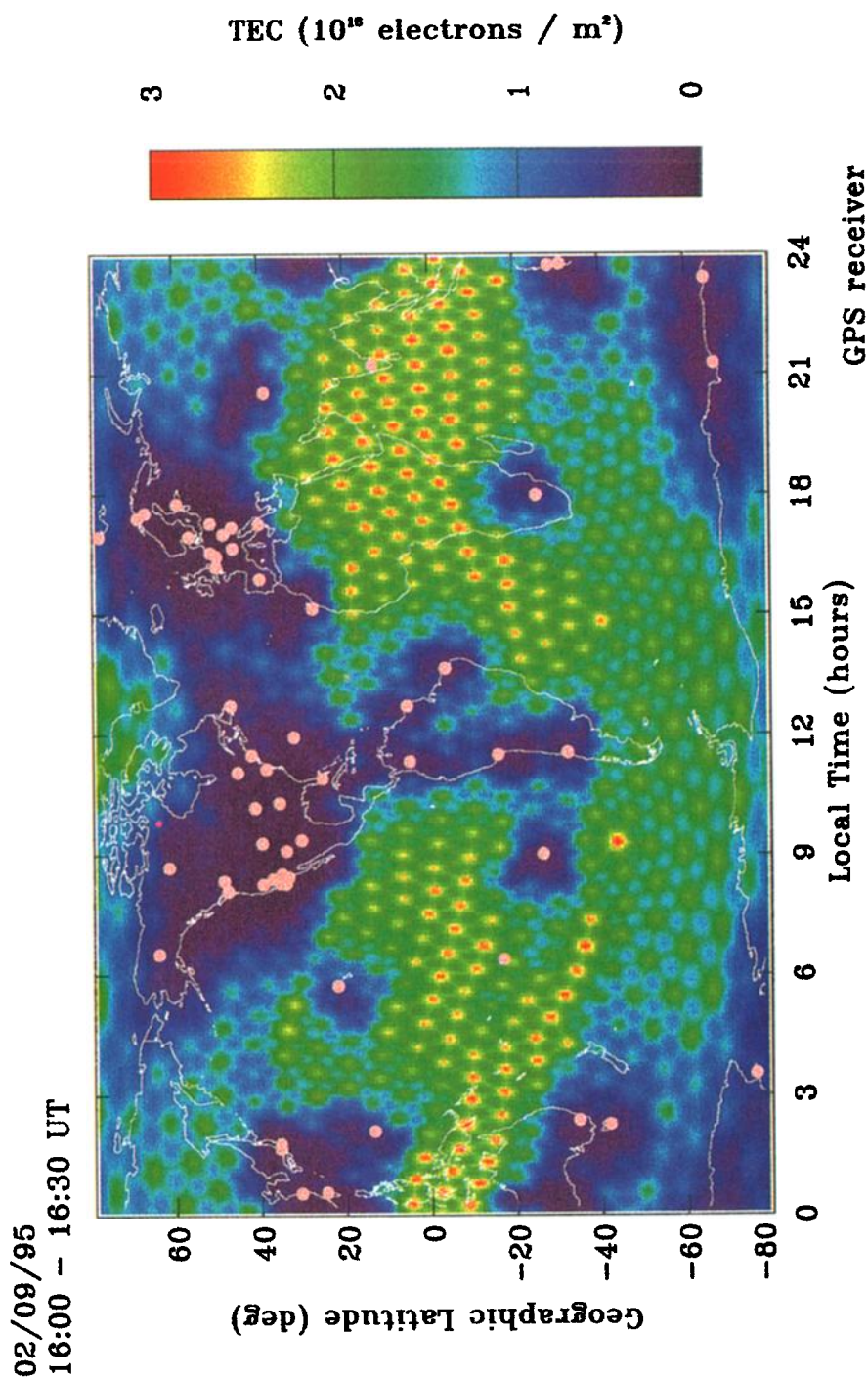


Plate 2. Formal error corresponding to the global map of Plate 1, scaled by a factor of 3.0 so that the low-latitude values reach a maximum of about 10 TECU (preliminary comparisons with TOPEX altimeter data indicate that this is roughly the accuracy of the maps in low latitude regions at large distances from the stations). These maps can be used for determining areas where the TEC retrieval is of reduced precision due to lack of measurements.

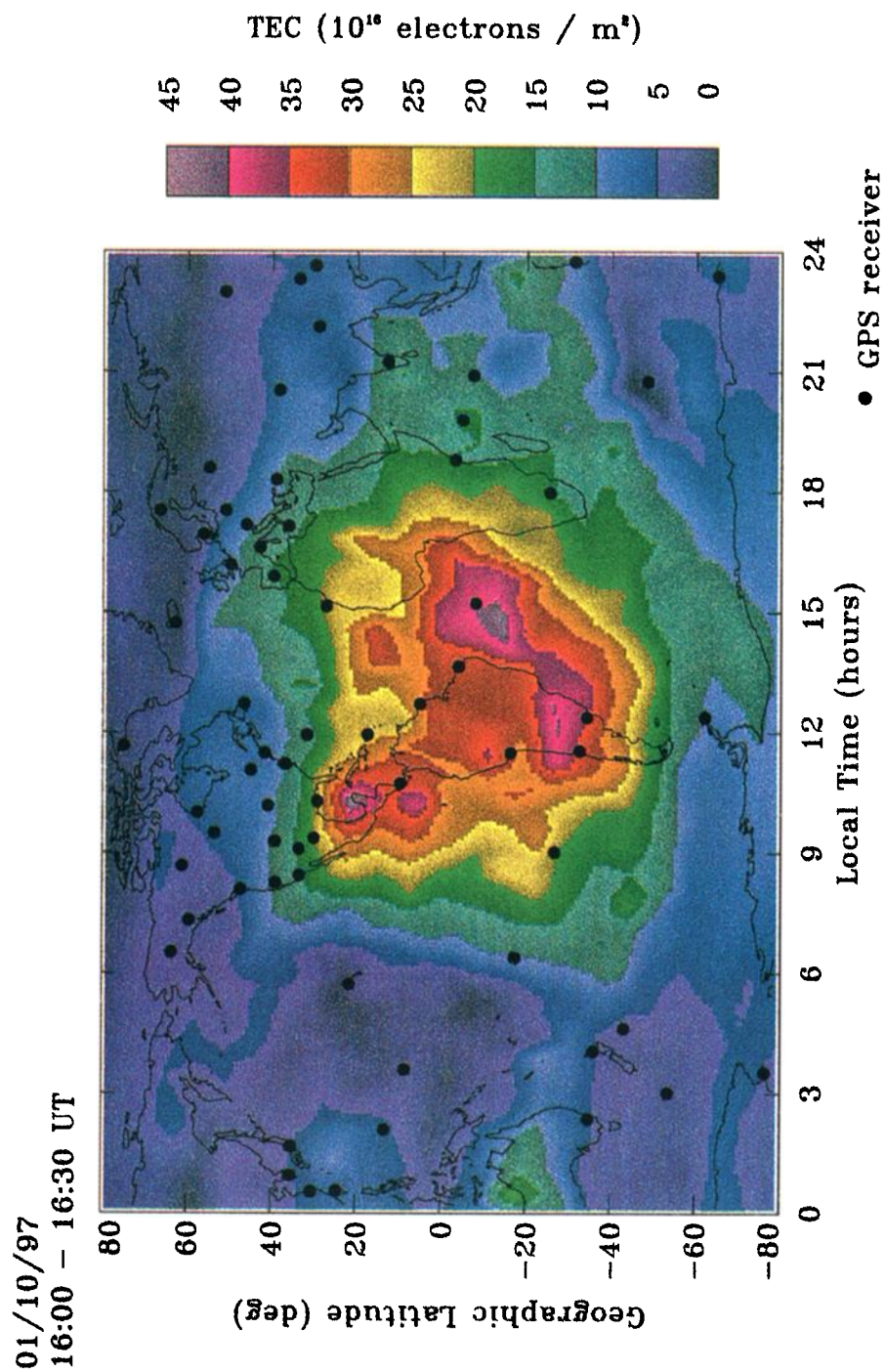


Plate 3. Global ionospheric TEC map during a geomagnetic storm that occurred on January 10, 1997.

geomagnetic latitude are similarly curved when projected onto the Earth's surface, due to the 11° offset between the Earth's rotational and geomagnetic dipole axes.

The formal error of the TEC retrieval is shown in Plate 2. The smallest errors occur in those continents containing clusters of closely spaced receivers (North America, Europe, and Australia), where the shell TEC distribution is directly sampled by measurements during the interval 1600–1630 UT. The TEC estimates for a particular tile vertex, which has a fixed local time, are based on current measurements when one or more receivers are in the vicinity of the vertex. At a later universal time, when the receivers shift to a different local time, the TEC estimate is based on the most recent direct measurements, although several hours may have elapsed since these were available. Mathematically, this persistence of the estimate is a consequence of the assumption that the TEC obeys a random walk in time: The estimated value is time independent. However, the formal error of the estimates increases with time (see section 3.4). This is why the formal error increases in areas at large distances ($> 1500\text{ km}$) from the nearest receiver.

A global TEC retrieval obtained during moderately disturbed geomagnetic conditions is shown in Plate 3 (January 10, 1997; daily $A_p = 32$). Many large-scale features are different than in Plate 1, which represents quiet geomagnetic conditions (daily $A_p = 3$). For example, in contrast to the quiet case where the nighttime TEC gradually decreases northward from 50° latitude, the disturbed case shows a TEC enhancement in the high-latitude (possibly auroral) region at night. In addition, there appears to be an enhancement at southern midlatitudes during local daytime (1300–1500 LT), which extends the low-latitude TEC bulge in the southern direction. The seasons of the global retrievals in Plates 1 and 3 are similar, so the southern enhancement may be storm related, possibly due to atmospheric disturbances such as composition changes or enhanced equatorward winds. This example suggests that TEC mapping based on global GPS data sets can be an important tool for retrieving features that appear during geomagnetically disturbed conditions [Ho *et al.*, 1996].

4.1. Assessing the Accuracy of Global TEC Maps

The global TEC retrievals have been compared with vertical TEC measurements obtained independently from the dual-frequency altimeter on board the TOPEX satellite [Callahan, 1993]. The use of two frequencies allows the radar altimetry data, obtained from an altitude of 1330 km, to be calibrated for ionospheric delay, which can be extracted separately with an estimated accuracy of about 3 TECU [Callahan, 1993].

The first comparisons between TOPEX and GPS TEC data were performed for the purpose of verifying the TOPEX altimeter ionospheric correction [Christensen *et al.*, 1994]. These comparisons occurred when the TOPEX satellite flew directly above a GPS receiver placed on the Harvest oil platform near the California coast (latitude 34.47° ; longitude 239.32°). Using satellite bias estimates derived from the global mapping technique, and a locally interpolated TEC map above the platform [Lanyi and Roth, 1988], 17 overflight comparisons indicate agreement at the 3.4-TECU level (RMS) (a hardware calibration technique was used to account for the receiver interfrequency bias). Thus it appears unlikely that the satellite bias estimates derived from the global mapping technique are determined to an accuracy significantly worse than 3 TECU.

We have also conducted comparisons to full tracks of TOPEX data, as they extend beyond the immediate proximity of a GPS receiver. These data cover a wide latitude range between $\sim -67^\circ$ and $+67^\circ$ and are available exclusively over the water mass, so comparisons to assess the accuracy of the global retrievals are also possible far from the GPS receivers. Such comparisons are useful for assessing the accuracy of the TEC interpolation over distances of several thousand kilometers in the solar-magnetic reference frame. Examples of TOPEX ground tracks are shown in Figure 5 for two daytime passes on February 9, 1995. The locations of GPS receivers are indicated as filled circles. The TOPEX orbit is nearly Sun-synchronous, so the local times of the measurements at a given latitude are nearly constant over a time span of a few days. In this case, local time varies from about 1000 at the northern edge, through

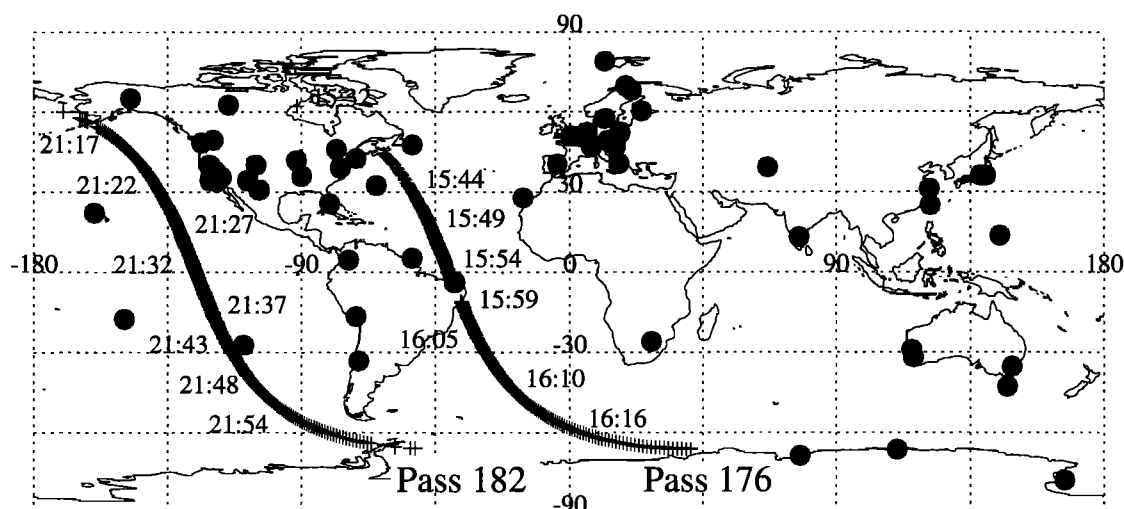


Figure 5. Location of TOPEX ground tracks (+) for passes 182 and 176 on February 9, 1995, and the location of GPS receivers in the global network (•).

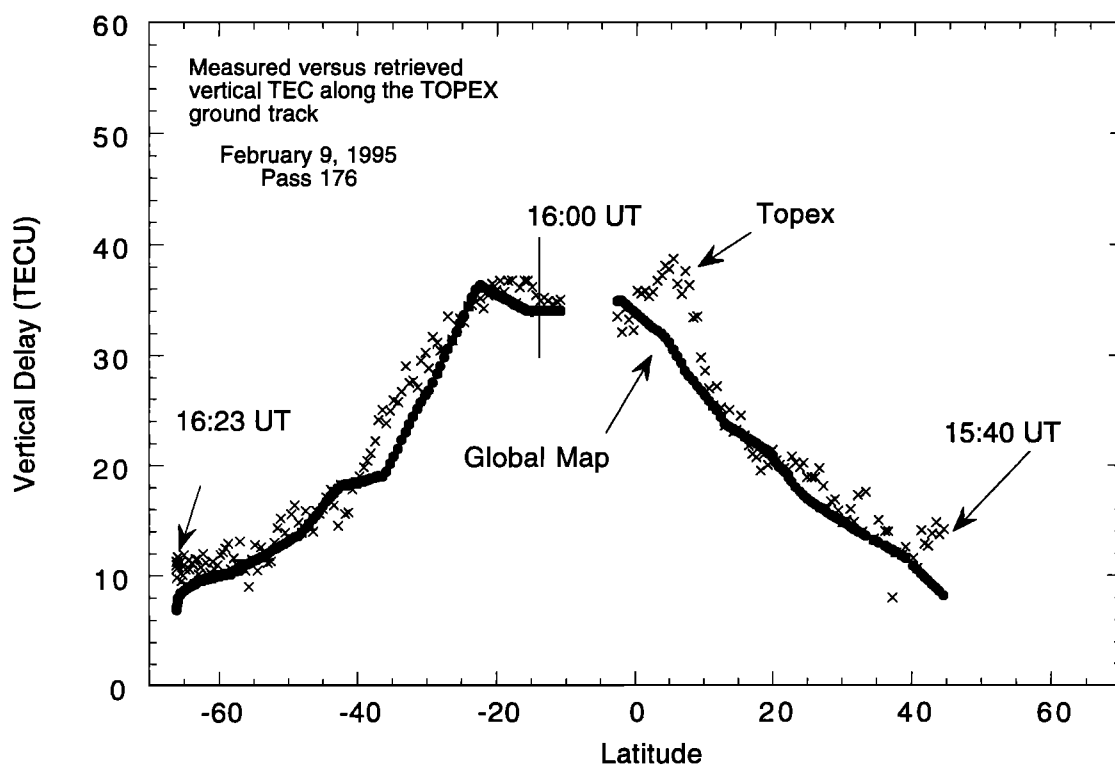


Figure 6. The TOPEX vertical TEC measurements, up to the spacecraft altitude of 1330 km, are plotted with the global map predictions along the TOPEX track, for pass 176 on February 9, 1995 (see Figure 5 for the location of the ground tracks).

1300 at the equator, to 1800 at the southern edge of the ground track.

A comparison between the TOPEX measurements and global ionospheric map retrievals is shown in Figure 6, plotted versus latitude for pass 176 on February 9, 1995. Successive maps were generated at 30-min intervals, so two map retrievals are necessary to generate the comparisons, where the time boundary is shown as a line in the figure. The maps follow the TEC measured by TOPEX quite accurately, over both low and middle latitudes. Referring to Figure 5, this track was located in the western edge of the Atlantic Ocean where data are available from the North and South American sectors. The square root of the random walk variance rate for this and all the other retrievals in this section was set to $0.8 \text{ TECU/h}^{1/2}$.

Another comparison between TOPEX and the GIM retrievals is shown in Figure 7, for pass 182 on February 9, 1995. In this case, TOPEX measures a significant TEC peak in the southern hemisphere. The large TEC gradient associated with this feature is

reproduced in the GIM retrieval fairly well. Referring to the geometry in Figure 5, the accuracy of the retrieval is correlated with the proximity of the GPS receivers: The southern low-latitude station at Easter Island is closer to the ground track than the northern low-latitude stations in South America. Accuracy improves markedly in the northern midlatitudes, where stations are closer and the ionospheric TEC is generally less variable.

The TEC along the TOPEX track predicted by the Bent model is also shown in Figure 7. This model was used to initialize the maps and was also used as a source of simulated data applied on an hourly basis. The weight of the model information was about 15 TECU, much less than the estimated noise of the GPS measurements, so it had a negligible effect on the global retrieval in the region of the ground track. The significantly improved agreement with TOPEX of GIM over Bent (see the mean and standard deviations of the predictions in the table in Figure 7) is often observed when comparisons have been made

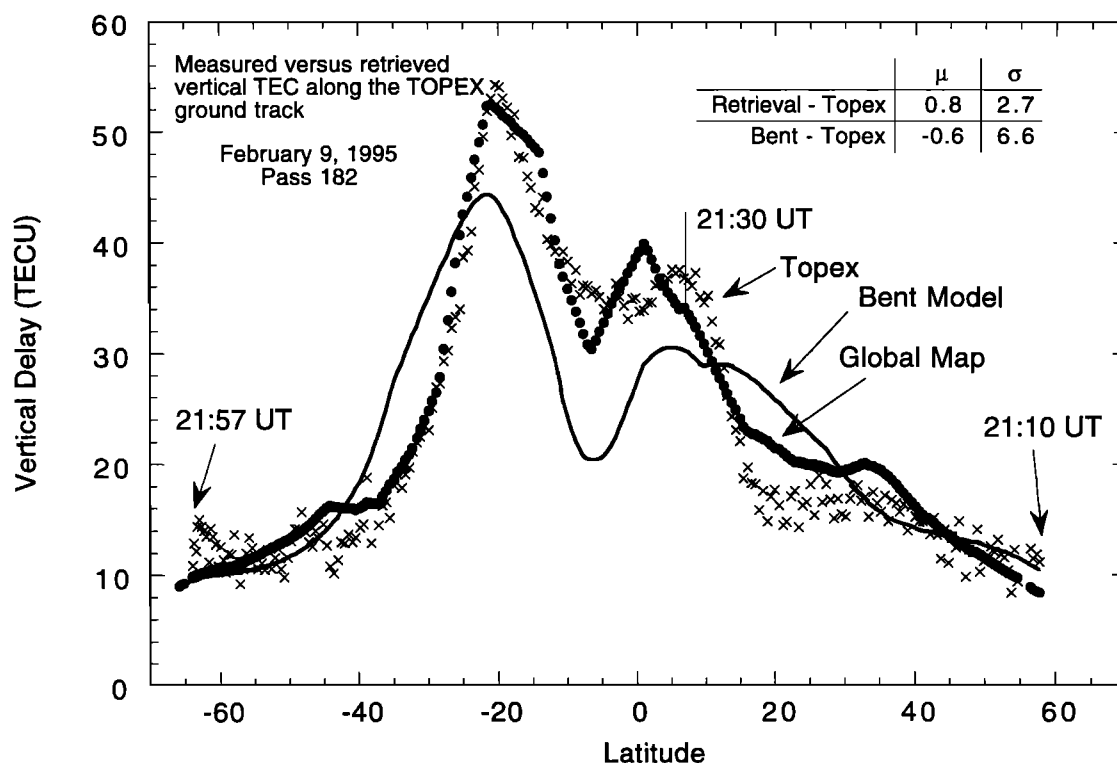


Figure 7. The TOPEX vertical TEC is compared with the global map retrieval for pass 182 on February 9, 1995. The prediction of the Bent climatological model for the TEC underneath TOPEX is also plotted. The mean difference (μ) and standard deviation of the difference (σ) between TOPEX and the two predictions is also shown.

between the data-driven mapping technique and climatological models (a detailed study is given by *Ho et al.*, 1997).

In these comparisons between GIM and TOPEX measurements, no adjustments have been applied to account for the plasmaspheric electron content above TOPEX. The good agreement obtained suggests that either TOPEX measurements have a compensating positive bias or the map values are too low. Electron content above 1330 km is often small (1-3 TECU); additional independent data of comparable accuracy may be necessary to detect the existence of such small systematic effects.

5. Summary

We have described a method for retrieving global vertical TEC distributions by interpolating GPS data from a global network of receivers. This method uses a shell model approximation (electron density concentrated at a fixed height) for assigning a unique shell intersect point to each slant measurement, and a fixed electron density profile for estimating vertical TEC. Spatial interpolation between the TEC measurements is performed with respect to a nearly Sun fixed triangular grid structure. Measurements within each triangular tile are used to estimate the TEC at the three tile vertices, assuming bilinear distance weighting. The approach is self-calibrating because receiver and transmitter instrumental biases can be estimated simultaneously with the TEC maps.

Weighted least squares fitting is used to perform the retrievals, using the Kalman filter algorithm. Updated maps are generated at subhourly intervals, and covariance information is fully propagated at each time step. The vertex TEC values are modeled as "random walk" stochastic parameters, that is, in the absence of new data, the TEC is constant with time (in the solar-magnetic reference frame) and the variance grows linearly with time.

The technique has been extended to include processing of simulated data from climatological models, so maps can be generated no matter how much data are available. Model data have been applied by simulating a vertical TEC measurement at each vertex of the triangular grid, at a fixed time interval of one hour, or initially before real data are processed. Of course, accuracy is highest when a well-distributed data set is available.

The TEC maps have been compared with vertical TEC measurements from the TOPEX dual-

frequency altimeter to determine whether spatial gradients observed in the ionosphere can be accurately retrieved. The TOPEX ground track covers a wide range of latitudes, including low latitudes where there are relatively few GPS receivers. These comparisons suggest that representing the global TEC distribution with the triangular gridding described here represents a reasonable model for reproducing latitudinal gradients even near the equator, although the wide range of possible equatorial TEC structures suggests that more extensive comparisons be performed. More comprehensive comparisons are currently in progress to determine how the accuracy of GIM retrievals degrades with distance from the stations [*Ho et al.*, 1997].

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